Mini Project Report

On

TRAFFIC LIGHT CONTROLLER USING 8051

Submitted in partial fulfillment of the requirements of

Mini Project III (ET 391)

Bachelor of Technology in Electronics and Telecommunication Engineering Prepared By:

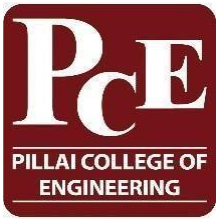
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CERTIFICATE

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Abstract

The Traffic Light Controller using the 8051 microcontroller is an embedded-system project designed to automate traffic signal operation for two lanes. The AT89C51 acts as the central controller and operates Red, Yellow, and Green LEDs in a fixed sequence. A two-digit common-anode seven-segment display provides a countdown for the active signal. The program uses Timer 0 for timing and multiplexes the two display digits rapidly so that the countdown can be shown clearly.

In the implemented sequence, Lane 1 receives a Green signal for 25 seconds followed by Yellow for 5 seconds, while Lane 2 remains Red. Control then changes to Lane 2, which receives Green for 25 seconds followed by Yellow for 5 seconds while Lane 1 remains Red. The cycle repeats continuously. The hardware uses an AT89C51, traffic LEDs, current-limiting resistors, a crystal oscillator, reset circuitry and two seven-segment displays.

The project demonstrates practical microcontroller programming, timer-based control, LED interfacing and multiplexed display operation. It provides a simple, low-cost model of an automatic traffic controller and can be extended in future with sensors for traffic-density-based control.

Keywords: 8051 Microcontroller, Traffic Light Controller, AT89C51, Timer 0, LEDs, Seven-Segment Display, Automation

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Summary

The Traffic Light Controller using the 8051 microcontroller is an automated two-lane traffic signal system. The main objective is to control vehicle movement using a predefined sequence of Red, Yellow and Green signals without manual intervention. The AT89C51 controls two traffic-light groups and a two-digit seven-segment countdown display.

Lane 1 is given Green for 25 seconds and Yellow for 5 seconds while Lane 2 remains Red. The sequence then moves to Lane 2, which receives Green for 25 seconds and Yellow for 5 seconds while Lane 1 remains Red. The countdown display shows the remaining time for each phase. Timer 0 is used to generate the basic 10 ms delay, and display multiplexing is used to drive the tens and ones digits.

The project combines hardware interfacing and embedded C programming. It demonstrates the use of AT89C51 I/O ports, timer operation, LED control and seven-segment display interfacing. The design can be extended with IR or ultrasonic sensors, adaptive timing, emergency-vehicle priority or IoT connectivity in future versions.

Chapter 01 Introduction

1.1 Fundamentals

Automation is widely used to improve efficiency, safety and reliability in transportation systems. Managing traffic flow at busy intersections is one of the major challenges in urban environments, and traffic signals are an important application because vehicles must be allowed to move in an organized sequence. Traditional manual control often leads to human error, uneven traffic distribution and congestion. Microcontroller-based controllers solve this by running a fixed, repeatable signal sequence automatically. Among the available microcontrollers, the 8051 is a popular and reliable choice because of its simplicity, versatility, multiple I/O ports, on-chip timers and ease of programming.

In this project, the 8051 microcontroller (AT89C51) is used as the main control unit for a two-lane traffic light controller. The controller operates Red, Yellow and Green LEDs according to programmed timing. A Green phase is provided for 25 seconds and a Yellow phase for 5 seconds. During the active phase of one lane, the other lane remains Red. A two-digit seven-segment display shows the countdown time. The project therefore demonstrates timing control, I/O interfacing and embedded C programming using the AT89C51, and provides a base that can later be upgraded with sensors for vehicle detection and adaptive control.

1.2 Objectives

1. To design and implement an automatic traffic light controller using the AT89C51 microcontroller.
2. To control Red, Yellow and Green LEDs for two lanes in a safe sequence.
3. To provide 25 seconds of Green time and 5 seconds of Yellow transition time for each lane.
4. To display the remaining time using a two-digit common-anode seven-segment display.
5. To use Timer 0 for accurate basic timing and multiplexed display operation.
6. To demonstrate practical hardware-software interfacing in an embedded system.
7. To develop a low-cost prototype that can be extended with sensors in the future.

1.3 Scope

The scope of the Traffic Light Controller covers the design and implementation of a small-scale automated traffic signal system using the 8051 microcontroller. The project focuses on fixed-time control, LED indication and countdown display. The project scope includes:

- Microcontroller-based automatic traffic signal control.
- Two-lane Red, Yellow and Green signal sequencing.
- 25-second Green and 5-second Yellow timing.
- Two-digit seven-segment countdown display using multiplexing.
- Timer 0 based delay generation.
- Hardware implementation using LEDs, resistors, oscillator and reset circuitry.
- A foundation for future sensor-based adaptive traffic control.

Future improvements can include vehicle-density sensing, adaptive signal timing, emergency-vehicle priority and wireless or IoT-based monitoring.

1.4 Outline

The report describes the hardware components, system operation, software development, circuit diagram, PCB implementation and final working of the Traffic Light Controller. Chapter 2 presents the hardware specifications, block diagram and prototype. Chapter 3 covers the software tools, flowchart and the embedded C program. Chapter 4 gives the circuit diagram and PCB layout with the fabrication process. Chapter 5 explains the working of the system and Chapter 6 concludes the report. The program is written in embedded C for the AT89C51 and was developed and simulated using standard embedded-system tools.

Chapter 02 Hardware Implementation

2.1 Specification of Hardware

1. AT89C51 Microcontroller

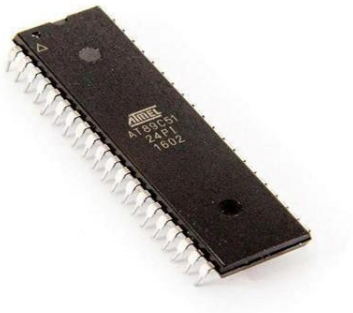


Fig. 2.1.1: 8051 Microcontroller

Fig. 2.1.1: 8051 Microcontroller

Specifications:

- 4K Bytes of In-System Reprogrammable Flash Memory – Endurance: 1,000 Write/Erase Cycles
- Fully Static Operation: 0 Hz to 24 MHz
- Three-level Program Memory Lock
- 128 x 8-bit Internal RAM
- 32 Programmable I/O Lines
- Two 16-bit Timer/Counters
- Six Interrupt Sources
- Programmable Serial Channel
- Low-power Idle and Power-down Modes

2. Seven Segment Display

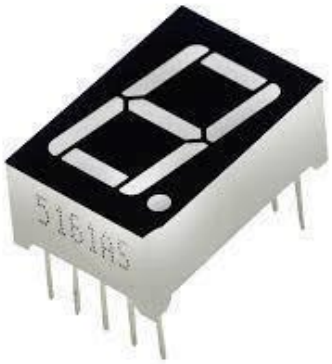


Fig. 2.1.2: Seven Segment Display

Specifications:

- Display Type: 7-segment LED display (common anode or common cathode); a common-anode type is used in this project
- Number of Segments: 7 segments (labeled a–g) and sometimes a decimal point (DP)
- Operating Voltage: Typically +3V to +5V DC
- Forward Current (per segment): Around 10–20 mA
- Power Dissipation: Approximately 60–150 mW (depending on color and current)
- Display Color: Commonly Red, but also available in Green, Yellow, or Blue
- Viewing Angle: Wide viewing angle, typically >60°
- Pin Configuration: 10 pins (2 for common connection, 8 for segments + DP)
- Material: Epoxy resin encapsulated plastic case
- Lifetime: Up to 100,000 hours of continuous operation

3. LED (Light Emitting Diode)



Fig. 2.1.3: LEDs

Specifications:

- Type: 5mm Round LED (used for Red, Yellow, and Green signals)
- Material: Gallium Arsenide Phosphide (GaAsP) or Gallium Phosphide (GaP)
- Operating Voltage: Typically 1.8V to 3.3V DC (varies by color)
- Red LED: ~1.8V – 2.0V; Yellow LED: ~2.0V – 2.2V; Green LED: ~2.0V – 3.0V
- Forward Current: 10–20 mA (typical operating range)
- Maximum Reverse Voltage: 5V
- Luminous Intensity: 100–2000 mcd (depending on color and size)
- Wavelength: Red ~630–660 nm, Yellow ~585–595 nm, Green ~520–530 nm
- Viewing Angle: 30° to 60° (standard LEDs)
- Power Dissipation: 50–100 mW per LED
- Lifetime: Up to 100,000 hours of continuous use

4. Other Components and Port Allocation

Table 2.1: Components Used in the System

Component	Quantity	Purpose
AT89C51 microcontroller	1	Central controller
Crystal oscillator with two 33 pF capacitors	1 + 2	Clock source for the 8051
Reset circuit (10 k-ohm, 10 uF, push button)	1 set	Power-on and manual reset
Red, Yellow, Green LEDs	2 each (6)	Signals for Lane 1 and Lane 2
1 k-ohm resistors	6	Current limiting for the LEDs
Common-anode seven-segment display	2	Tens and ones digit of countdow

n

Table 2.2: Port and Pin Allocation of the AT89C51

Port / Pin	Function	Logic
P1.0 – P1.2	Lane 1 Red, Yellow, Green LEDs	0 = ON
P1.3 – P1.5	Lane 2 Red, Yellow, Green LEDs	0 = ON
P0 (P0.0 – P0.7)	Segment data (a–g, dp) of the display	0 = segment ON
P2.0	Tens digit select	1 = enabled
P2.1	Ones digit select	1 = enabled

2.2 Block Diagram of the System

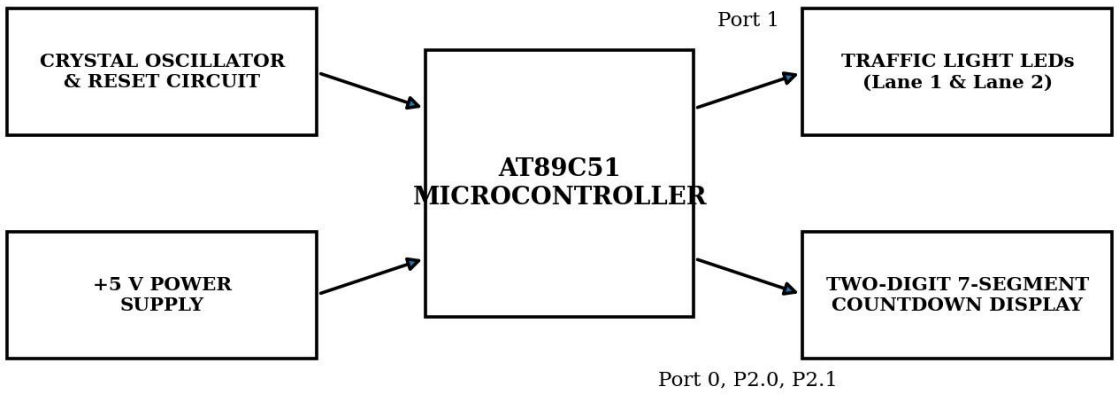


Fig. 2.2.1: Block diagram of the system

The block diagram represents the interconnection of the AT89C51 microcontroller, traffic LEDs and the two-digit seven-segment countdown display. The microcontroller receives the programmed control sequence and generates the required logic signals for both traffic-light groups. The crystal oscillator and reset circuit provide the clock and start-up reset, while the +5 V supply powers all parts of the circuit.

The Red LED indicates that vehicles must stop, the Yellow LED indicates a transition or warning period, and the Green LED allows vehicles to move. The seven-segment display shows the remaining time of the active phase. Timer 0 provides the basic timing interval while Port 0 supplies segment data and Port 2 selects the display digits.

The system operates in a continuous sequence: Lane 1 Green, then Lane 1 Yellow, then Lane 2 Green, then Lane 2 Yellow, and then repeats.

2.3 Breadboard / Simulation Prototyping

The design was first verified as a simulation prototype in Proteus before the PCB was fabricated. The simulated circuit with the AT89C51, the two sets of traffic LEDs and the two seven-segment displays is shown below.

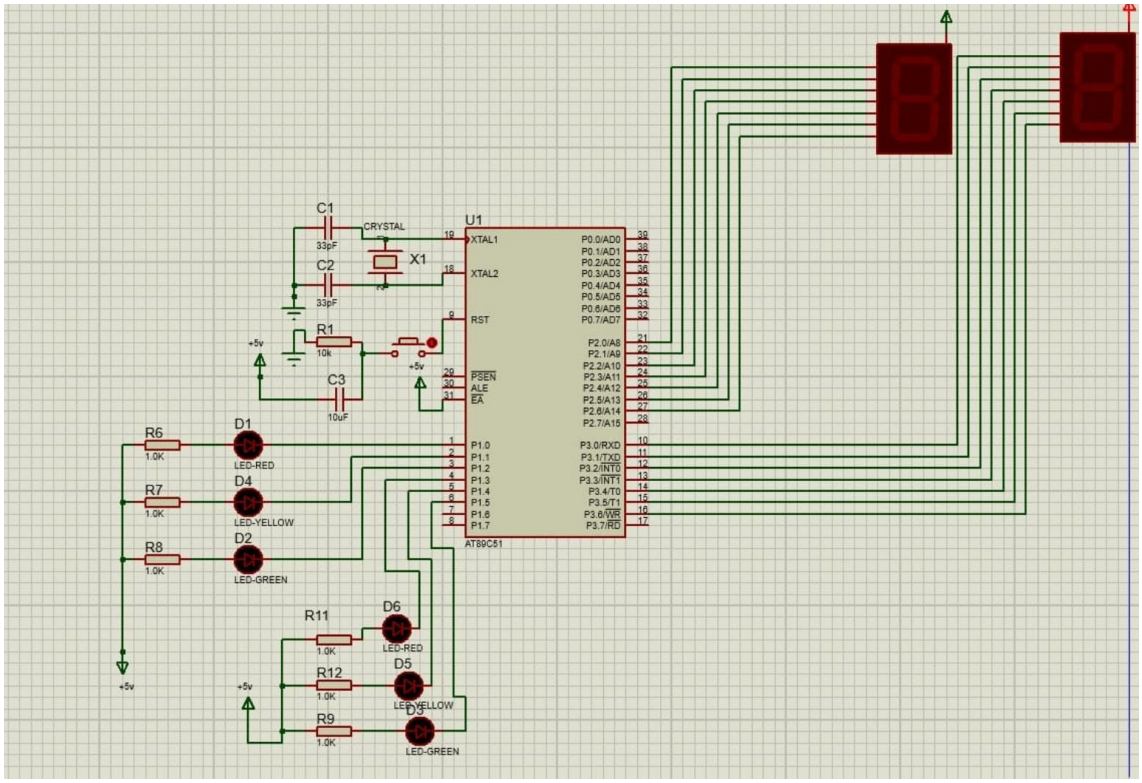


Fig. 2.3.1: Simulation prototype of the system (Proteus)

Chapter 3 Software Implementation

3.1 Overview

3.1.1 Keil uVision 5

Keil uVision 5 is a powerful integrated development environment (IDE) widely used for developing and testing embedded applications. It provides a range of features that help write, compile, and debug embedded programs. One of its key features is support for Software Components: the Run-Time Environment (RTE) in MDK Version 5 allows validated software components to be used in a project, which greatly speeds up development and helps ensure that the code is reliable and efficient.

Keil uVision 5 also provides comprehensive documentation and context-sensitive help for API functions, along with code templates that help in starting quickly and serve as a reference. In addition, it offers improved support for real-time operating systems (RTOS) with features like stack watermarking, which is useful for complex embedded systems that handle multiple tasks simultaneously.

3.1.2 Proteus 8 Professional

Proteus 8 Professional is an advanced software tool used for designing and simulating electronic circuits. It provides a complete platform where users can create circuit diagrams, simulate their operation, and even design printed circuit boards (PCBs). It is highly popular among electronics students, hobbyists, and professionals because it allows real-time testing of circuits without requiring physical components, saving both time and cost during development.

One of its main features is the ability to simulate microcontroller-based projects. It supports a wide range of microcontrollers such as 8051, PIC, AVR, ARM, and Arduino, and includes virtual instruments like oscilloscopes, voltmeters, and logic analyzers that help observe circuit behavior during simulation. Once the schematic is verified through simulation, the PCB layout can be designed directly, ensuring an efficient workflow from concept to implementation.

3.2 Flowchart

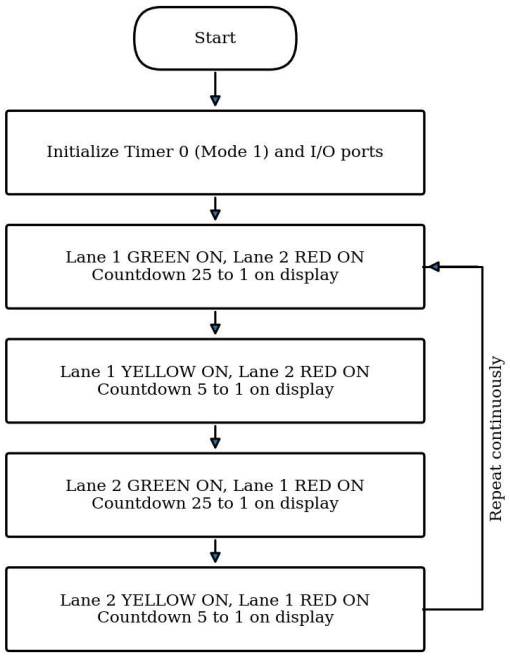


Fig. 3.2.1: Flowchart of the system

- **Start** – The process begins with powering on the system.
- **Initialize** – Timer 0 is set to Mode 1 (16-bit) and the I/O ports are configured for the LEDs and the seven-segment display.
- **Lane 1 Green** – Lane 1 Green and Lane 2 Red are turned ON. The display counts down from 25 to 1.
- **Lane 1 Yellow** – Lane 1 Yellow and Lane 2 Red are ON. The display counts down from 5 to 1.
- **Lane 2 Green** – Lane 2 Green and Lane 1 Red are ON. The display counts down from 25 to 1.
- **Lane 2 Yellow** – Lane 2 Yellow and Lane 1 Red are ON. The display counts down from 5 to 1.
- **Repeat** – The controller returns to Lane 1 Green and the cycle repeats continuously.

3.3 Program / Code

```
#include <reg51.h>

/* ---------------- TRAFFIC LIGHTS (Port 1) ---------------- */
sbit L1_RED    = P1^0; sbit L1_YELLOW = P1^1; sbit L1_GREEN  =
P1^2;

sbit L2_RED    = P1^3;
sbit L2_YELLOW = P1^4;
sbit L2_GREEN  = P1^5;

/* ---------------- DIGIT SELECTION (Port 2) ---------------- */
sbit DIGIT1_TENS = P2^0; sbit DIGIT2_ONES = P2^1;

/* ---------------- COMMON ANODE LOOKUP TABLE ---------------- */
// 0 = Segment ON, 1 = Segment OFF unsigned char code SEG[10] = {
0xC0, /* 0 */
    0xF9, /* 1 */
    0xA4, /* 2 */
    0xB0, /* 3 */
    0x99, /* 4 */
    0x92, /* 5 */
    0x82, /* 6 */
    0xF8, /* 7 */
    0x80, /* 8 */
    0x90  /* 9 */
};

/* ---------------- TIMER 0 DELAY (10 ms) ---------------- */
void delay_10ms(void) {     TH0 = 0xDC;
    TL0 = 0x00;
TR0 = 1;
while(TF0 == 0);
TR0 = 0;
    TF0 = 0;
}

/* ---------------- MULTIPLEXED DISPLAY (1 Second Duration) ---------------- */
// Drives both displays rapidly for 1 second without flickering or garbling
void display_1sec(unsigned char number) {     unsigned char tens = number / 10;
unsigned char ones = number % 10;     unsigned char i;

    // Loop 50 times (50 x 20ms = 1000ms = 1 second)
for(i = 0; i < 50; i++) {         // --- Display
Tens Digit ---
        P0 = SEG[tens];       // Send Common Anode code to Port 0
DIGIT1_TENS = 1;      // Enable Tens Digit (for BC547 NPN Driver)
```

```
DIGIT2_ONES = 0;      // Disable Ones Digit         delay_10ms();
// Wait 10ms

        P0 = 0xFF;            // Clear Port 0 to eliminate ghosting
        DIGIT1_TENS = 0;

        // --- Display Ones Digit ---
        P0 = SEG[ones];       // Send Common Anode code to Port 0
        DIGIT1_TENS = 0;      // Disable Tens Digit         DIGIT2_ONES =
1;      // Enable Ones Digit (for BC547 NPN Driver)         delay_10ms();
// Wait 10ms
        P0 = 0xFF;            // Clear Port 0 to eliminate ghosting
        DIGIT2_ONES = 0;
    }
}

/* ---------------- TRAFFIC PHASES ---------------- */ void
lane1_green(void) {     unsigned char t;     L1_RED = 1; L1_YELLOW =
1; L1_GREEN = 0;  // Lane 1 Green (0 = ON)     L2_RED = 0; L2_YELLOW =
1; L2_GREEN = 1;  // Lane 2 Red   (0 = ON)

    for(t = 25; t > 0; t--) {
display_1sec(t);     } }

void lane1_yellow(void) {     unsigned char t;     L1_RED = 1;
L1_YELLOW = 0; L1_GREEN = 1;  // Lane 1 Yellow     L2_RED = 0;
L2_YELLOW = 1; L2_GREEN = 1;  // Lane 2 Red

    for(t = 5; t > 0; t--) {
display_1sec(t);     } }

void lane2_green(void) {     unsigned char t;     L1_RED = 0;
L1_YELLOW = 1; L1_GREEN = 1;  // Lane 1 Red     L2_RED = 1;
L2_YELLOW = 1; L2_GREEN = 0;  // Lane 2 Green

    for(t = 25; t > 0; t--) {
display_1sec(t);     } }

void lane2_yellow(void) {     unsigned char t;     L1_RED = 0;
L1_YELLOW = 1; L1_GREEN = 1;  // Lane 1 Red     L2_RED = 1;
L2_YELLOW = 0; L2_GREEN = 1;  // Lane 2 Yellow

    for(t = 5; t > 0; t--) {
display_1sec(t);     }
}

/* ---------------- MAIN ---------------- */
void main(void) {     TMOD = 0x01;   // Timer
0, Mode 1 (16-bit)
    TR0 = 0;
TF0 = 0;

    while(1) {
lane1_green();
lane1_yellow();
lane2_green();
lane2_yellow();     }
}
```

Chapter 4 Circuit Diagram and PCB Layout

4.1 Circuit Diagram

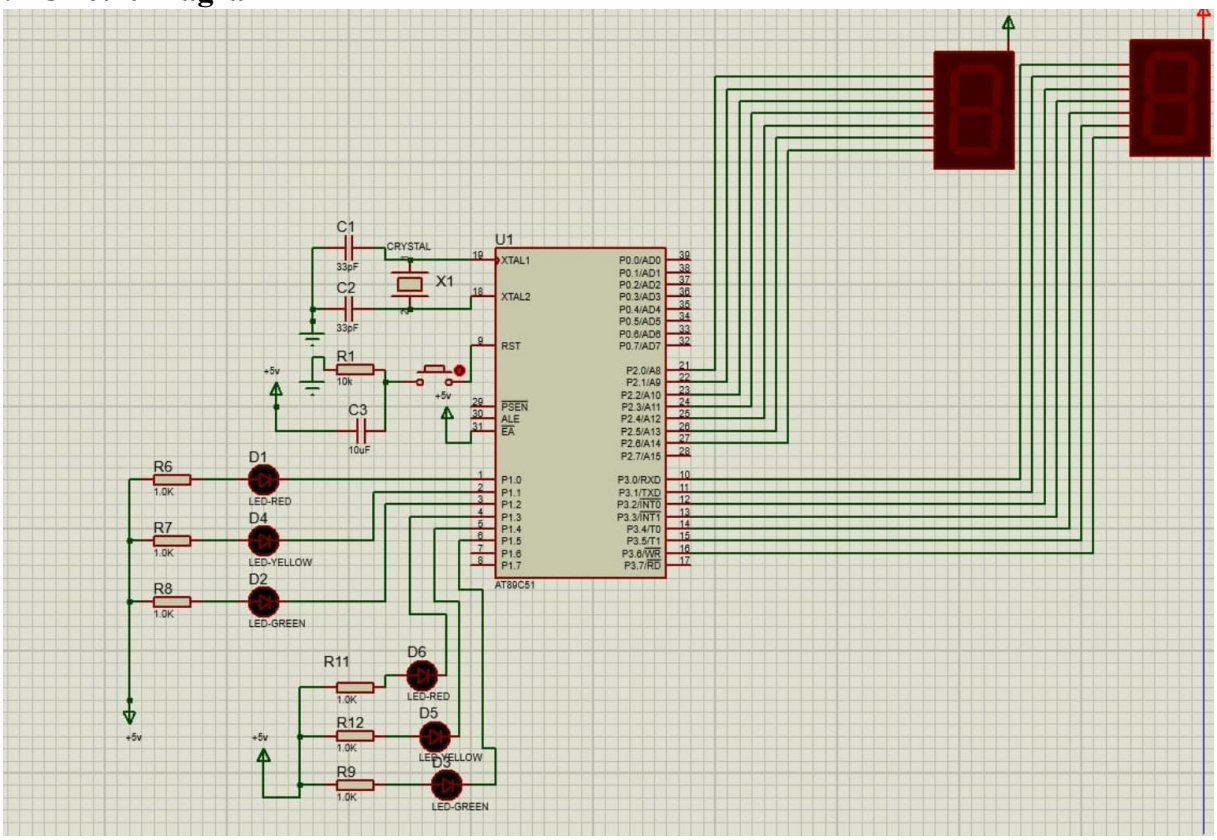


Fig. 4.1.1: Circuit diagram of the Traffic Light Controller

The circuit is built around the AT89C51 (U1) with a crystal (X1) and two 33 pF capacitors (C1, C2) on XTAL1/XTAL2. The reset circuit consists of a 10 k-ohm resistor (R1), a 10 uF capacitor (C3) and a push button. The traffic LEDs (Red, Yellow, Green for each lane) are connected to Port 1 through 1 k-ohm current-limiting resistors, and the two seven-segment displays form the countdown output.

4.2 PCB Layout

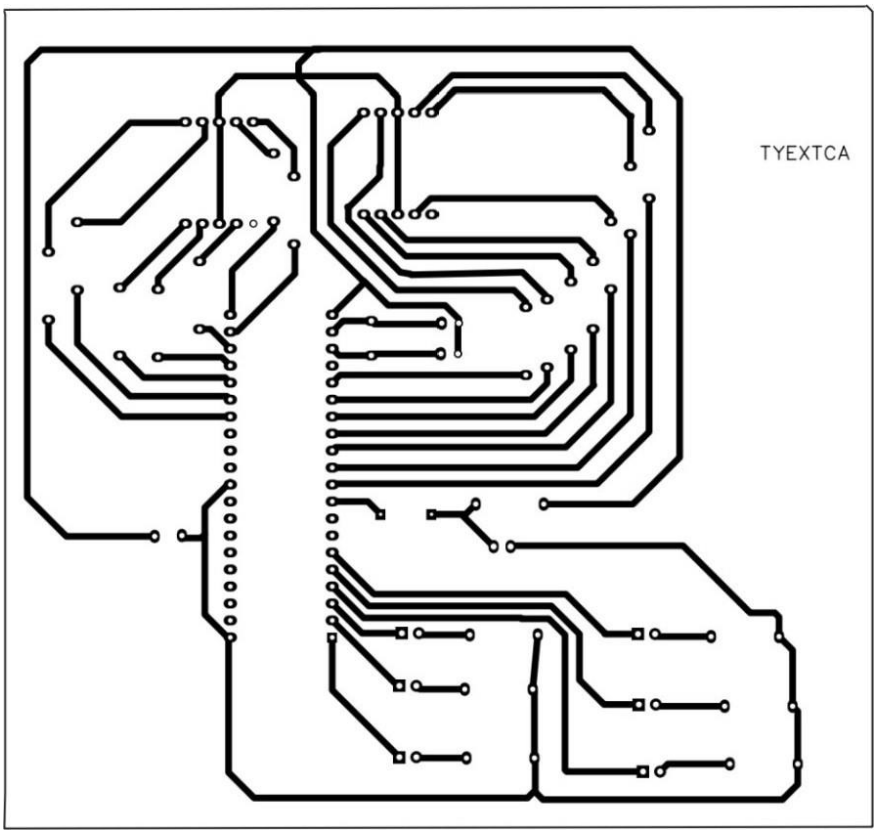


Fig. 4.2.1: Artwork of PCB layout

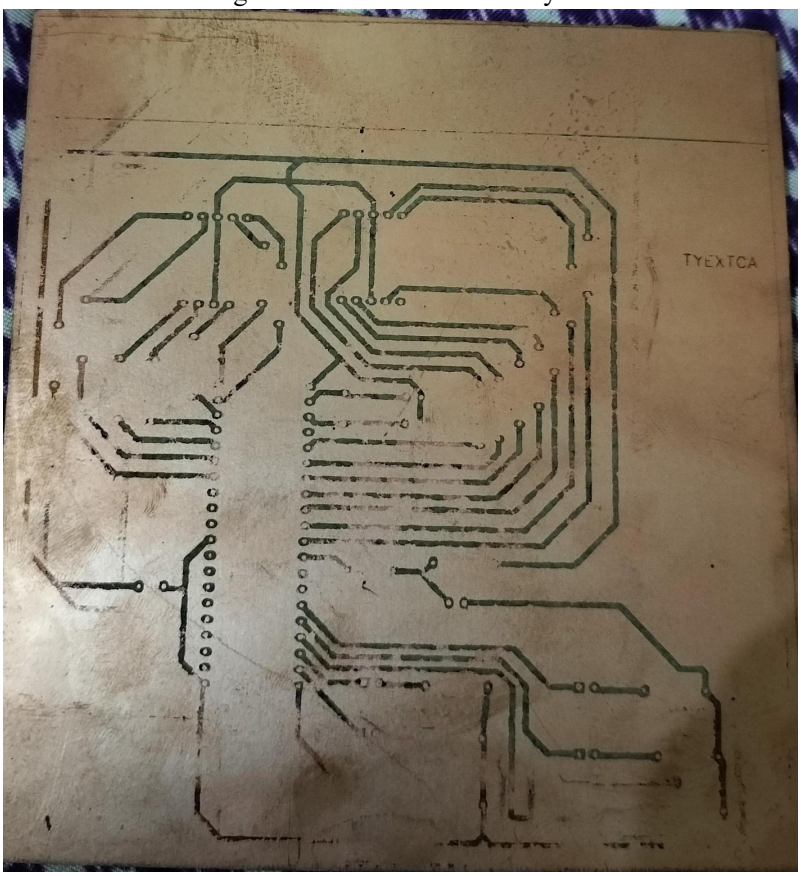


Fig. 4.2.2: PCB layout after etching process

PCB Fabrication Process

During the development of the project, the process began with designing the PCB layout. After completing the design, the layout was printed on photo paper for further processing. The copper-clad board was prepared by thoroughly cleaning and scrubbing its surface to ensure it was free from impurities. The printed layout was then transferred onto the copper board by carefully ironing the photo paper over it to imprint the circuit design.

Once the transfer was complete, the board was inspected for any discontinuities or broken tracks, which were corrected using a permanent marker. The board then underwent the etching process, where unwanted copper was removed, leaving behind the precise circuit pattern. After etching, a continuity test was performed to identify and fix any broken connections to ensure the circuit's reliability.

Following this, holes were drilled at specific locations to mount the electronic components. The components were then soldered onto the PCB in accordance with the circuit layout. Finally, the assembled PCB underwent functional testing to verify its proper operation. This systematic procedure ensured the successful fabrication of the PCB and validated the accuracy and effectiveness of both the design and assembly processes.

Chapter 05 Working

The Traffic Light Controller is based on the AT89C51 microcontroller and controls two traffic-light groups and a two-digit countdown display. The working process is explained below.

1. **Initialization:** When the system is powered on, the microcontroller initializes Timer 0 (Mode 1, 16-bit) and the I/O ports connected to the LEDs and the seven-segment display.
2. **Lane 1 Green Phase:** Lane 1 Green and Lane 2 Red are ON for 25 seconds. Vehicles in Lane 1 move while the display counts down from 25 to 1.
3. **Lane 1 Yellow Phase:** Lane 1 Yellow is ON and Lane 2 remains Red for 5 seconds. This warns Lane 1 vehicles to prepare to stop.
4. **Lane 2 Green Phase:** Lane 2 Green and Lane 1 Red are ON for 25 seconds, with the display counting down from 25 to 1.
5. **Lane 2 Yellow Phase:** Lane 2 Yellow is ON and Lane 1 remains Red for 5 seconds.
6. **Display Operation:** The common-anode seven-segment codes are selected from the SEG lookup table. The tens and ones digits are multiplexed rapidly (10 ms each) to produce a stable two-digit countdown, and Port 0 is cleared between digits to avoid ghosting.
7. **Cycle Repetition:** After Lane 2 Yellow, the controller returns to Lane 1 Green and repeats continuously without manual intervention.
8. **Timing Control:** Timer 0 provides the basic 10 ms delay. The display routine repeats 50 multiplexing cycles of 20 ms each, which gives the one-second interval for every countdown step.

Table 5.1: Signal Sequence and Timing of One Cycle

Phase	Lane 1	Lane 2	Duration	Display
1	Green	Red	25 s	25 to 1
2	Yellow	Red	5 s	5 to 1
3	Red	Green	25 s	25 to 1
4	Red	Yellow	5 s	5 to 1

One complete cycle therefore takes 60 seconds, after which the sequence restarts from Phase 1.

Chapter 06 Conclusion

The Traffic Light Controller using the AT89C51 microcontroller has been successfully designed as a two-lane automatic traffic signal system. The project demonstrates sequential control of Red, Yellow and Green LEDs along with a two-digit seven-segment countdown display.

The controller provides 25 seconds of Green time and 5 seconds of Yellow time for each lane. While one lane is active, the other lane remains Red, providing a simple and systematic traffic-control sequence. Timer 0 is used for timing and the display is multiplexed to show the countdown clearly.

The project provides practical experience in microcontroller programming, I/O interfacing, timer operation, LED control and PCB implementation. It also provides a foundation for future

improvements such as traffic-density sensors, adaptive timing, emergency-vehicle priority and IoT-based monitoring.